Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/18/2025
PI Name:	Paul, Anna-Lisa Ph.D.		
Project Title:	Epigenetic Adaptation to the Spaceflight Environment - Accumulated Genomic Change Induced by Generations in Space		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential (2) Bioregenerative Life Support		
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Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	10/26/2018	**End Date:**	10/25/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/25/2024 per NSSC information (Ed., 2/7/24) NOTE: End date changed to 10/25/2023 per NSSC information (Ed., 7/29/23) NOTE: End date changed to 10/25/2022 per NSSC information (Ed., 9/17/21)		
Key Personnel Changes/Previous PI:	Addition of Assistant Research Scientist Dr. Minqi Zhou to the project with expertise in epigenetic regulation in plants. Addition of Postdoctoral researcher Dr. Vicken Aknadibossian to the project with expertise in epigenetics		
COI Name (Institution):	Barbazuk, William Ph.D. (University of Florida, Gainesville) Ferl, Robert Ph.D. (University of Florida, Gainesville)		
Grant/Contract No.:	80NSSC19K0130		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Changes in the epigenome of plants, specifically DNA methylation patterns, are now known to occur in spaceflight. These changes are only beginning to be understood as far as their effect on physiological adaptation and productivity in spaceflight; however, the fact that DNA methylations are altered by spaceflight is clear, and disruption of methylation mechanisms affects growth in space. What is not yet understood, or even approached, is the trans-generational stability of these epigenetic changes, and whether establishing spaceflight methylation patterns within seeds can improve next-generation adaptability to spaceflight. Therefore, this proposal seeks to examine the transmission of DNA methylation patterns through seeds produced in plants grown on the International Space Station (ISS). Seeds produced on orbit will be used for second generation of spaceflight growth. This second ISS grow out, comprising seeds from spaceflight-grown plants together with seeds from terrestrially-grown plants, will determine whether second generation epigenetic changes continue to accumulate or stabilize. The second grow out will also reveal any spaceflight-associated changes in plant growth and development or gene regulation that are epigenetically regulated between plants grown from seeds produced on the ISS and seeds produced on the Earth. This experiment will determine whether growing a generation of plants in space confers an adaptive advantage to the next generation, similar to epigenetic adaptive advantage known to occur with some terrestrial plant stresses. This project is one of both discovery on a genome-wide scale and practical application to exploration needs. We propose unbiased whole genome bisulfite sequencing in Arabidopsis leaves across spaceflight generations to discover epigenetic modification patterns over generations, coupled with RNASeq examination of the comparable transcriptomes to assess spaceflight-associated DNA methylation on transcript profiles across generations. This will greatly advance fundamental understanding of the role of epigenetics in spaceflight adaptation. The practical applications arise from the fact that epigenetic imprinting occurs in plants exposed to environmental stress. Acquired DNA methylation patterns can transfer to the next generation, where they contribute to the ability of the progeny to cope with the stress experienced by the parental generation. Should that occur in spaceflight, we will have identified genetic elements that increase the adaptability of plants to spaceflight -- the first steps in developing spaceflight cultivars bred to better serve the roles of plants in life support. The tactical objectives of this proposal are: 1) Bisulfite whole genome sequencing to map spaceflight-induced changes in 5mCyt at single base-pair resolution for leaves of mature plants, to enrich the methylome data that we currently have for seedling leaves and roots. 2) RNASeq to assess transcriptome in the same samples, as differential methylation is linked to changes in gene expression that characterize the functional impacts of spaceflight across the entire genome in maturing leaves. 3) Collecting seeds produced on orbit and growing a second generation on orbit. The second flight grow out will be used to gather growth, development, gene expression, and epigenetic data on second generation plants on orbit. During both growth cycles on orbit, images of plant growth will monitor development. Elucidating multigenerational spaceflight methylomes contributes to the fundamental understanding of how biology responds and adapts to the spaceflight environment, and provides insight in the adaptive strategies plants use in the face of novel stimuli outside of their evolutionary experience. The long-term goal of this line of inquiry is to understand how the methylome contributes to the adaptive strategies of plants to spaceflight in order to breed plants that are specifically and better adapted to spaceflight in microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research contributes to a greater understanding of the role of epigenetic changes to the genome in response to novel or stressful environments. Further, it will be the first experiment to test whether the epigenomic changes observed in plants grown in a spaceflight environment can be inherited to the next generation, and confer an adaptive advantage to spaceflight physiology.
Task Progress:	The returned samples from PH-03A and PH-03B were processed for seed germination tests, RNA and DNA extractions, and processing for subsequent sequencing. For reference in the text below, the Advanced Plant Habitat (APH) growth area was divided into four quadrants. All quadrants of PH-03A were planted with terrestrially-grown seed. In PH-03B, the International Space Station (ISS) Flight (FLT) grown seed was planted in the even-named quadrants (2,4) and the terrestrially-grown Ground Control (GC) seed planted in the odd-numbered (1,3) quadrants. The results are not yet published, so only approaches are indicated in this public section. Seed germination tests: The Harvested seed from PH-03A (equal mass from each quadrant combined) from FLT and from GC were used to replant diagonal quadrants for each treatment. In PH-03B, both FLT and GC planted as: FLT-A-source in odd Quadrants, GC-A-source in Even Quadrants. Written below as: in PH03B (Q1,3) = 03A FLT seed; (Q2,4) = 03A GC seed. Transcriptome and Methylome comparisons: In PH03B (Q1,3) = 03A FLT seed; (Q2,4) = 03A GC seed. 1 – 03A FLT to 03A GC (All Quads) 2 – 03B FLT(Q1,3) to 03B GC(Q1,3) 3 – 03B FLT(Q2,4) to 03B GC(Q2,4) 4 – 03B FLT(Q1,3) to 03B FLT(Q2,4) 5 – 03B GC(Q1,3) to 03B GC(Q2,4) 6 – 03A FLT(Q1,3) to 03B FLT(Q1,3) 7 – 03A FLT(Q2,4) to 03B FLT(Q2,4) 8 – 03A GC(Q1,3) to 03B GC(Q1,3) 9 – 03A GC(Q2,4) to 03B GC(Q2,4) 10 – 03A FLT(Q1,Q3) to 03A FLT(Q2,Q4) 11 – 03B FLT(Q1,Q3) to 03A GC(All Quads) 12 – 03B FLT(Q2, Q4) to 03A GC(All Quads) Manuscripts are in preparation and are likely to make it through the submission and review process for publication in mid to late 2025.
Bibliography Type:	Description: (Last Updated: 11/20/2025)
Articles in Peer-reviewed Journals	Zhou M, Riva A, Gauthier ML, Kladde MP, Ferl RJ, Paul AL. "Single-molecule long-read methylation profiling reveals regional DNA methylation regulated by Elongator Complex Subunit 2 in Arabidopsis roots experiencing spaceflight." Biol Direct. 2024 Apr 30;19(1):33. https://doi.org/10.1186/s13062-024-00476-z ; PMID: 38689301; PMCID: PMC11059628. [NOTE: previously reported in November 2022 as "on preprint server".] , Apr-2024

Articles in Peer-reviewed Journals	Olanrewaju GO, Haveman NJ, Naldrett MJ, Paul A-L, Ferl RJ, Wyatt SE. "Integrative transcriptomics and proteomics profiling of Arabidopsis thaliana elucidates novel mechanisms underlying spaceflight adaptation." Front Plant Sci. 2023 Nov 27;14:1260429. https://doi.org/10.3389/fpls.2023.1260429 ; PMID: 38089794; PMCID: PMC10712242 , Nov-2023